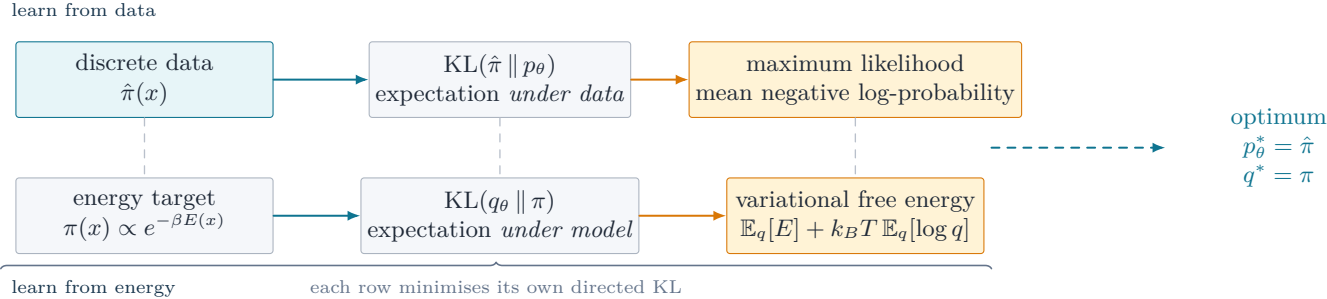
learn from data
discrete data
$\hat{\pi}(x)$
$\mathrm{KL}(\hat{\pi} \,\|\, p_\theta)$
expectation *under data*
maximum likelihood
mean negative log-probability
optimum
$p_\theta^* = \hat{\pi}$
$q^* = \pi$
energy target
$\pi(x) \propto e^{-\beta E(x)}$
$\mathrm{KL}(q_\theta \,\|\, \pi)$
expectation *under model*
variational free energy
$\mathbb{E}_q[E] + k_B T\, \mathbb{E}_q[\log q]$
learn from energy
each row minimises its own directed KL